

TEC Heating and Cooling Analysis: Open-Loop Calibration and the Slope Ratio

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We measured the steady-state temperature of a thermoelectric cooler (TEC, Laird CP14-127-045-L2-W4.5) driven by signed PWM through an H-bridge. Both branches are linear, with slopes $m_h = 0.485^\circ\text{C}/\text{count}$ (heating) and $m_c = 0.172^\circ\text{C}/\text{count}$ (cooling), giving $r = m_h/m_c = 2.82$. A steady-state energy balance with PWM-averaged current gives $\dot{Q}_J/\dot{Q}_P = (r - 1)/(r + 1) = 0.48$. The Laird maximum-current data predict $r_{\text{Laird,max}} = 2.56$. We discuss why the two ratios need not agree.

1. PART 4: COMBINED GRAPH AND FITS

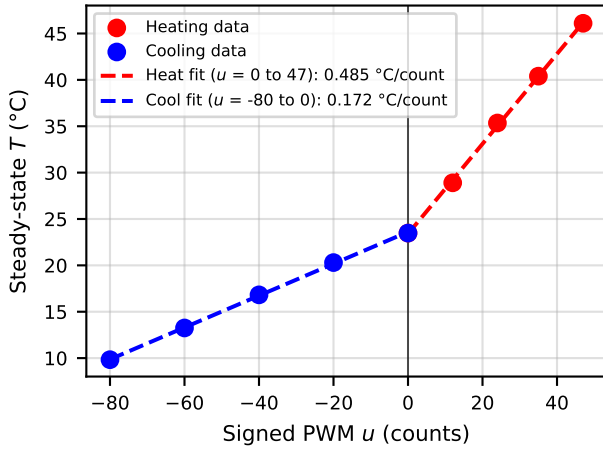


FIG. 1. Steady-state temperature versus signed PWM u (positive = heating, red; negative = cooling, blue), with a least-squares line for each branch. Heating fit: $u = 0$ to 47; cooling fit: $u = -80$ to 0. Both branches share the $u = 0$ point (23.48°C). Steady state: after each PWM step we waited about three time constants ($\tau \lesssim 100\text{s}$), then recorded the temperature once it stayed within $\pm 0.1^\circ\text{C}$ (about the normal measurement noise) for one further minute. Waits were 370–725 s.

2. PART 5.1: MEASURED SLOPES

Linear fits to the five points of each branch in Fig. 1 give

$$m_h = 0.485 \pm 0.007^\circ\text{C}/\text{count} \quad (u = 0 \text{ to } 47), \quad (1)$$

$$m_c = 0.172 \pm 0.002^\circ\text{C}/\text{count} \quad (u = -80 \text{ to } 0), \quad (2)$$

$$r = \frac{m_h}{m_c} = \frac{0.485}{0.172} = 2.82 \pm 0.05, \quad (3)$$

where the uncertainties are the standard errors of the fitted slopes. The duty ranges are $D \leq 0.184$

(heating) and $D \leq 0.314$ (cooling). Neither branch shows visible curvature: the slopes between neighbouring points are 0.45, 0.54, 0.46, 0.48 (heating) and 0.17, 0.18, 0.17, 0.16 $^\circ\text{C}/\text{count}$ (cooling), with no systematic trend. We therefore fitted one straight line to all five points of each branch, from 0 to that direction's maximum useful PWM.

3. PART 5.2: PWM AND THE SLOPE-RATIO MODEL

Steady-state balance. At steady state $G(T - T_0) = \dot{Q}_{\text{TEC}}$. $\dot{Q}_{\text{TEC}}$ holds the two current-dependent object-face terms: Peltier transport (into the object when heating, out of it when cooling) and object-face Joule heating $\frac{1}{2}I^2 R_M$ (always into the object). $G(T - T_0)$ is the net passive heat flow out of the object, including conduction back through the TEC. When heating ($T > T_0$) it flows out; when cooling ($T < T_0$) it flows in. The individual flows are not zero; their sum is.

PWM averages. In one period τ the current is I for $D\tau$ and 0 for $(1 - D)\tau$, so

$$\langle I \rangle = \frac{1}{\tau} \left[\int_0^{D\tau} I dt + \int_{D\tau}^\tau 0 dt \right] = DI, \quad (4)$$

$$\langle I^2 \rangle = \frac{1}{\tau} \int_0^{D\tau} I^2 dt = DI^2. \quad (5)$$

Because the current is either full or zero, $\langle I^2 \rangle = DI^2 \neq \langle I \rangle^2 = D^2 I^2$ for $0 < D < 1$. Both terms are therefore linear in D , and the model predicts a straight line on each branch with a kink at $u = 0$, as our graph shows. With $\langle I^2 \rangle = D^2 I^2$ (as for DC), the heating branch would curve upward and the cooling branch would flatten. This assumes the PWM period ($\approx 2\text{ms}$) is much shorter than the thermal time constant, so the block responds only to the averages.

Slope ratio. With signed duty $d = u/255$ and $D = |d|$, $\dot{Q}_{\text{TEC}} = d\dot{Q}_P + |d|\dot{Q}_J$. For heating ($d > 0$, $|d| = d$) and cooling ($d < 0$, $|d| = -d$), the steady-state balance

gives

$$T_h - T_0 = \frac{d(\dot{Q}_P + \dot{Q}_J)}{G}, \quad \frac{dT_h}{dd} = \frac{\dot{Q}_P + \dot{Q}_J}{G}, \quad (6)$$

$$T_c - T_0 = \frac{d(\dot{Q}_P - \dot{Q}_J)}{G}, \quad \frac{dT_c}{dd} = \frac{\dot{Q}_P - \dot{Q}_J}{G}. \quad (7)$$

Both slopes are positive when $\dot{Q}_P > \dot{Q}_J$, and the heating slope is larger. Since $m = dT/du = \frac{1}{255} dT/dd$, the factors $1/255$ and G cancel:

$$r = \frac{m_h}{m_c} = \frac{\dot{Q}_P + \dot{Q}_J}{\dot{Q}_P - \dot{Q}_J}. \quad (8)$$

Let $x = \dot{Q}_J/\dot{Q}_P$. Then $r = (1+x)/(1-x)$, so $r - rx = 1+x$ and $r-1 = x(r+1)$:

$$\frac{\dot{Q}_J}{\dot{Q}_P} = \frac{r-1}{r+1} = \frac{2.82-1}{2.82+1} = 0.48 \pm 0.01. \quad (9)$$

(Check: $r = 2$ gives $1/3$.) At our on-state current, the object-face Joule heat is about half of the Peltier heat pumping.

4. PART 5.3: LAIRD DATA-SHEET CALCULATION

Table I lists the values from the Laird CP14-127-045-L2-W4.5 data sheet (Rev. 00, 06-01-2022), p. 3, *Specifications* table, hot-side temperature $T_h = 27.0^\circ\text{C}$ column.

TABLE I. Laird data-sheet values at $T_h = 27^\circ\text{C}$.

Quantity	Value	Meaning and condition
R_M	1.50 Ω	Module electrical resistance.
$I_{\max}$	8.6 A	Current giving $\Delta T_{\max}$; $Q_c = 0$.
$Q_{c,\max}$	71.3 W	Max cold-side heat; $I_{\max}$, $\Delta T = 0$.
$\Delta T_{\max}$	70.5 $^\circ\text{C}$	Max hot-to-cold difference; $I_{\max}$, $Q_c = 0$.

At $\Delta T = 0$ the conduction term is zero, and half of the Joule heat goes to the object face:

$$\dot{Q}_{J,\max} = \frac{1}{2} I_{\max}^2 R_M = \frac{1}{2} (8.6 \text{ A})^2 (1.50 \Omega) = 55.5 \text{ W}, \quad (10)$$

$$\dot{Q}_{P,\max} = Q_{c,\max} + \dot{Q}_{J,\max} = 71.3 + 55.5 = 126.8 \text{ W}, \quad (11)$$

$$r_{\text{Laird},\max} = \frac{\dot{Q}_{P,\max} + \dot{Q}_{J,\max}}{\dot{Q}_{P,\max} - \dot{Q}_{J,\max}} = \frac{182.3}{71.3} = 2.56. \quad (12)$$

At $I_{\max}$ this corresponds to $\dot{Q}_J/\dot{Q}_P = 55.5/126.8 = 0.44$.

5. PART 5.4: COMPARE THE RATIOS

Measured: $r = 2.82$ ($\dot{Q}_J/\dot{Q}_P = 0.48$). Laird at $I_{\max}$: $r = 2.56$ (0.44). The measured ratio is about 10% higher, which is similar but not equal. Exact agreement is not expected:

- **$D = 1$ does not mean $I = I_{\max}$.** Full duty only keeps the H-bridge on. The on-state current is set by the 12 V supply, the H-bridge and wiring drops, R_M , and the TEC's Seebeck back-voltage. That gives at most $12 \text{ V}/1.50 \Omega \approx 8 \text{ A}$ and in practice less, below $I_{\max} = 8.6 \text{ A}$. The 10 A supply limit is not reached.
- **The Joule fraction scales with current:** $\dot{Q}_J/\dot{Q}_P = \frac{1}{2} I^2 R_M / (S_M T_o I) \propto I$. A lower current alone would make r smaller than 2.56, so the excess comes from the effects below.
- **Peltier term $\propto T_o$ (kelvin).** The object face averages $\approx 308 \text{ K}$ on the heating branch and $\approx 290 \text{ K}$ on the cooling branch, so Peltier pumping is relatively stronger when heating. This raises r .
- **Finite ΔT , PWM, changing properties.** The data sheet is for steady DC at $\Delta T = 0$. Our object face spans 10 – 46°C , which changes TEC conduction and the back-voltage. R_M itself rises from 1.50 to 1.68Ω between 27 and 50°C (same table), and S_M and K_M also vary.
- **Passive paths and fitting.** If the leaks are not equal above and below room temperature, G does not cancel exactly. One straight line per branch also averages over any slight curvature.

6. PART 5.4: PASSIVE CONDUCTION

When the object is hotter than its surroundings (heating), passive heat flows *out of* the object; when it is colder (cooling), passive heat flows *into* it. Either way it pulls the temperature back toward T_0 , so it opposes both heating and cooling. Approximately symmetric conduction enters both branches through the same G . It reduces both slopes by the same factor $1/G$, which cancels from r . Without Joule heating the two branches would be $\pm \dot{Q}_P/G$, equal in magnitude. The unequal slopes come from Joule heating, which keeps its sign when the current reverses: it adds to Peltier heating and subtracts from Peltier cooling.